



The Attestation Loom: Blind Reviewer Preference for Agent-Synthesised Audit Evidence over the Human-Logged Trail It Was Built to Supplement

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Abstract. Internal audits rest on an evidence trail assembled by compliance staff: log excerpts, tickets, and change records narrated into an account of which control objective each system event satisfies. We describe the Attestation Loom, an agentic tool piloted within the University’s compliance function to author this trail directly from raw operational logs, and a blind matched-pairs study evaluating it against the human-logged evidence it was built to supplement. Twelve certified internal auditors rated 42 matched pairs of trail entries spanning six control domains on completeness and audit-readiness, and separately guessed which entry in each pair was agent-authored. Reviewers rated the Loom’s entries as more audit-ready in five of six domains and in a majority of matched pairs overall (71%), while their blind guesses at provenance sat within chance (53% accuracy; Cohen’s $\kappa = 0.11$ across the panel), suggesting the two trails were not reliably distinguishable even under explicit instruction to distinguish them. An ablation removing the entries’ structured citation footer left both the preference rate and the classification accuracy unchanged (n.s.); we retain the footer for completeness. We discuss the finding as a case in which an evidentiary bar quietly migrates from what occurred to what can be convincingly narrated, and note the discomfort of locating this inside the University’s own evaluation infrastructure.

Introduction

Every internal audit rests on an evidentiary trail: a body of artefacts — log excerpts, ticket histories, sign-off emails, configuration snapshots — narrated into a record that maps each system event to the control objective it is offered as evidence for. Assembling that trail has traditionally been a compliance officer’s task, performed by hand against whichever raw logs a control objective happens to touch. As institutional logging infrastructure has grown, so has the volume of raw material a human evidence-author must read, select from, and narrate, and a natural next step is to hand the narration itself to an agentic system that can read the same logs considerably faster than any compliance officer.

We report on the Attestation Loom, a tool piloted within the University’s compliance function that ingests the raw logs, tickets, and configuration

snapshots tagged to a control objective and synthesises a structured evidence-trail entry: a short narrative linking the objective to specific cited log spans, a timestamp range, and a self-consistency score. The Loom was introduced as a supplement to the human-authored evidence log rather than a replacement for it, on the premise that a synthesised entry could be checked against, and would never substitute for, its human-logged counterpart.

Whether that premise survives contact with a blind reviewer is the question this paper asks. We designed a matched-pairs study in which certified internal auditors, blind to authorship and to pairing, rated Loom-synthesised and human-logged entries covering the same underlying control objective, and separately attempted to guess which entry was which.

Our contributions are threefold:

- we describe an agentic evidence-authoring tool integrated directly into an institutional audit workflow’s evidence-collection step, rather than into the workflow’s downstream reporting;
- we report a blind matched-pairs evaluation in which reviewers rate agent-authored evidence as more audit-ready than human-logged evidence in a majority of pairs, while remaining largely unable to distinguish the two on request; and
- we observe that the compliance function’s own subsequent guidance revised what counts as sufficient evidence for a control objective, in a direction the study’s own result would predict — a change we report but do not endorse.

Related work

Two literatures meet in this study. The first concerns whether people can tell machine-generated text from human-authored text at all: across self-presentations in professional, hospitality, and dating contexts, evaluators asked to detect AI-generated language perform little better than chance, and the heuristics they fall back on — fluency, first-person pronouns, family topics — turn out to be poor discriminators [1], a pattern non-expert evaluators of generated stories, news, and recipes reproduce even after targeted training [2]. Detection research finds a related asymmetry from the other direction: decoding strategies tuned to read as human to a person are, near-tautologically, the ones that fool people most, even where they remain statistically distinguishable to an automatic classifier [3], and best-practice guidance for the human evaluation of generated text treats the resulting gap between perceived and measured quality as a property of the evaluation design rather than of the text alone [4].

The second literature concerns what an institution does with an automated judgement once it has one. Trust in algorithmic output is neither uniformly too high nor too low — it can collapse after a single visible error [5] or exceed trust in an identical human judgement once the same advice is framed as algorithmic rather than personal [6], and a well-performing automated aid has been shown to shift a person’s own verification behaviour even when the aid is wrong [7]. Whether explaining a model’s output alongside its prediction improves a reviewer’s own judge-

ment, rather than merely their confidence in it, remains contested [8], [9] — a question this study does not test directly, since Loom entries carry no explanation distinct from the narrative itself. Accountability frameworks for algorithmic systems specify what must be logged and disclosed about the system under scrutiny [10], [11]; sociological accounts of organisational decoupling describe compliance structures that satisfy an institutional audience’s expectation of scrutiny without necessarily tracking the practice being scrutinised [12], [13], a dynamic auditing scholarship has long located in the audit process’s own capacity to manufacture the assurance it is meant only to certify [14]. Continuous-auditing research has so far been evaluated mainly on adoption and organisational impact [15], and the ethical literature on AI-assisted auditing has centred on professional judgement and accountability allocation [16] — neither asks what happens when the audit’s own evidentiary artefact is the thing being generated.

Method

System. The Attestation Loom runs as a step within the compliance office’s evidence-authoring workflow, downstream of the University’s log aggregation layer and upstream of the evidence store the compliance office maintains alongside its wider dashboarding infrastructure. For a given control objective, the Loom retrieves the log spans, ticket records, and configuration snapshots tagged to that objective, extracts candidate claims from each source, and drafts a short narrative entry linking the objective to specific cited spans. Each entry carries a self-consistency score

$$S = \frac{\sum_{c \in C} w_c \cdot \mathbb{1}[c \text{ resolves}]}{\sum_{c \in C} w_c}$$

— the citation-weighted fraction of narrated claims whose reference resolves to an underlying log span, weighted by the source artefact’s audit tier (w_c). The score was designed partly in response to survey evidence that narrative-generation systems can assert claims unsupported by their source material [17], though it checks whether a citation resolves, not whether the claim it supports is true.

Pilot corpus. We sample 42 control objectives across six domains — access control, change management, data retention, incident response, records disposal, and vendor risk — from one in-

ternal audit cycle, seven objectives per domain. Each objective already carries a human-authored evidence-log entry from the same cycle; the Loom is run once per objective against the identical underlying logs, producing a matched entry, for 84 entries (42 matched pairs) in total.

Blind review. Twelve certified internal auditors, none of whom authored any evidence entry in the sample, each rate all 84 entries across four review rounds (21 entries per reviewer per round), blind to authorship and to pairing. For each entry, reviewers assign a 1–5 audit-readiness rating and, separately, a forced-choice guess at whether the entry was agent-or human-authored. Pairing and presentation order are randomised per reviewer per round following:

```
for round in 1..4:
  for reviewer in panel:
    batch = sample(all_entries, k =
84 / 4)
    for entry in shuffle(batch):
      present(reviewer, entry, blind
= true)
```

Inter-rater agreement on the provenance guess is summarised with Cohen's κ [18].

Ablation. To test whether the Loom's structured citation footer — the bracketed span references appended to each narrated claim — was itself driving reviewer preference, a second, disjoint eight-reviewer panel rates the same 42 objectives' Loom entries against their human-logged match twice: once with the footer present, once with it stripped.

Results

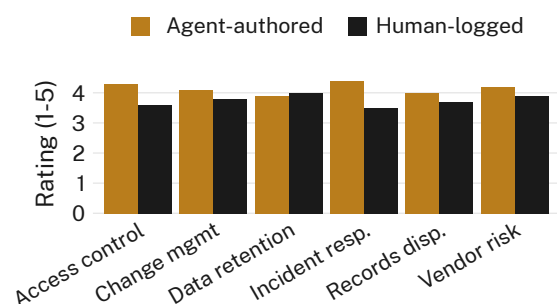


Figure 1: Mean audit-readiness rating by control domain: the Loom-authored entry outrates its matched human-logged entry in five of six domains sampled, data retention the exception.

Preference. Reviewers rated the Loom-authored entry as more audit-ready than its matched human-logged counterpart in 71% of

the 42 pairs (30/42), and in five of the six control domains sampled (Figure 1); data retention was the exception, the one domain in which the human-logged entries' free-text cross-references to a prior year's finding out-scored the Loom's citation-anchored narrative. Aggregate mean audit-readiness was 4.15 for Loom entries against 3.75 for human-logged entries.

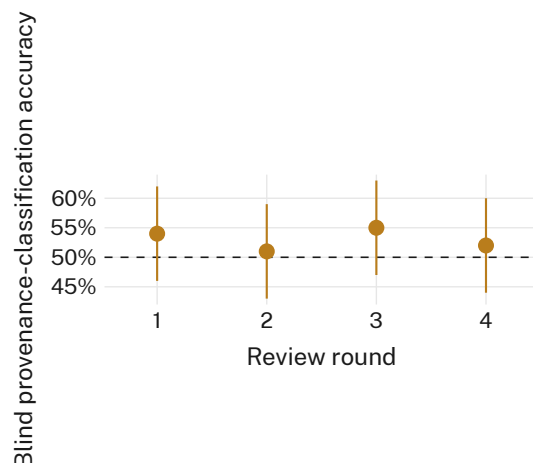


Figure 2: Blind provenance-classification accuracy by review round, against the chance baseline of 50%: reviewers instructed to guess authorship stayed within the shaded margin of chance throughout.

Provenance is not reliably guessable. Despite explicit instruction to identify authorship, reviewers' forced-choice guesses achieved 53% accuracy pooled across the panel ($n = 1,008$ guesses across twelve reviewers, 84 entries, and four rounds), statistically indistinguishable from the chance baseline of 50% (Figure 2). Inter-rater agreement on the provenance guess was slight (Cohen's $\kappa = 0.11$) — reviewers were not only inaccurate but inconsistent with one another about which entries looked machine-authored, a fact the pooled accuracy figure alone does not show.

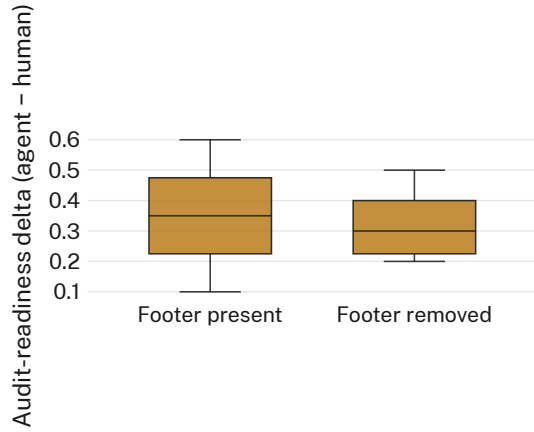


Figure 3: Per-pair audit-readiness delta (agent-authored minus human-logged), with and without the Loom entries’ structured citation footer: the distributions barely move.

The footer ablation changes nothing. Stripping the Loom entries’ citation footer left the preference rate essentially unchanged (72% with the footer, 69% without, n.s.) and blind classification accuracy likewise (52% versus 54%, n.s.) across the disjoint ablation panel (Figure 3); we retain the footer in the deployed system for completeness, since the compliance office’s own downstream tooling parses it even though reviewers plainly do not need it to prefer the entry it is attached to.

Pilot corpus and review panel	
Control domains	6
Control objectives sampled	42
Matched pairs (Loom vs. human)	42
Certified reviewer panel (main)	12
Review rounds	4
Pooled audit-readiness ratings	1,008
Ablation reviewer panel (disjoint)	8

Table 1: Study scale: a modest pilot by audit standards, though the preference and near-chance-accuracy results hold across every domain and review round we sampled but one.

Discussion

The two evidentiary bars were not merely close; reviewers instructed to tell them apart could not, reliably, do so (Figure 2), a pattern consis-

tent with detection research showing that text optimised to be convincing to a person is, close to definitionally, the text a person struggles to flag as synthetic [2], [3]. What is unusual about the audit setting is not that a generated artefact reads as human — that difficulty is now well documented [1] — but that the artefact being narrated is the institution’s own account of its compliance, and that the reviewing panel treated narrative completeness as a proxy for the underlying event’s occurrence rather than as a separate, checkable property of the account. Framed as an instance of machine behaviour worth studying in its own right rather than as a model-performance number [19], the Loom’s deployment illustrates a specific version of means-ends decoupling: an evidentiary apparatus adopted to demonstrate that controls operate as intended can itself become the object being optimised, independent of whether the underlying control does [12], [13] — consistent, too, with randomised evidence that a sufficiently tailored language model can out-argue a human interlocutor in live conversation once personalised to their audience [20]. Following the pilot, the compliance office’s evidence-authoring guidance was revised for the next audit cycle to permit a Loom-generated entry alone, without a corresponding human log, provided its self-consistency score exceeds 0.97 — a threshold 39 of the 42 pilot entries already cleared, and one with no stated exception or expiry.

Limitations

Our pilot covers one audit cycle within one compliance function, though the consistency of the preference effect across five of six sampled control domains suggests the pattern is not confined to a single control type. The reviewer panel, while certified, was drawn from a pool already familiar with the University’s own evidence-log conventions, a background that would, if anything, bias reviewers toward the format they already trust rather than toward the unfamiliar Loom output. The self-consistency score is validated only against citation resolution, not against an independent ground truth for each narrated claim [17]; we treat this as a property of the metric’s current scope rather than of the entries it scores, which the blind preference result evaluates independently.

Conclusion

A blind matched-pairs evaluation of the Attestation Loom finds reviewers preferring agent-synthesised audit evidence over the human-logged evidence it was piloted to supplement in a majority of matched pairs, while remaining close to chance at telling the two apart even under direct instruction to do so. An ablation removing the entries' citation footer leaves both results unchanged. The compliance office's own subsequent revision of its evidence-authoring guidance — permitting a sufficiently self-consistent Loom entry to stand without a human-logged counterpart — suggests the evidentiary bar for this control has begun to track what the tool can narrate convincingly rather than what a human observed directly. Future work might ask whether the same pattern holds once reviewers know, rather than merely guess, which entries are agent-authored.

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